

Task 1A: Initial Data Analysis

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April 21, 2025

Date 2025-04-21 — *Git commit:* `db8377f`

Overview

This document summarises the steps completed for all tasks - firstly, 1A - in the data science project. The focus has been on understanding how perceived wine quality varies between red and white varieties, through data analysis: specifically, distribution plots.

Quality Distribution – Red Wine

The first analysis targeted the distribution of the `quality` variable in the red wine dataset. The quality values range from 3 to 8, with most samples concentrated around 5 and 6.

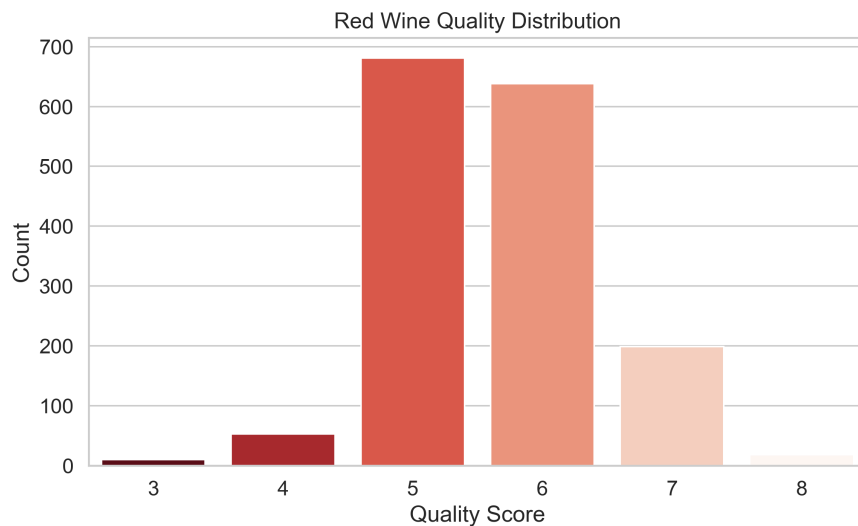


Figure 1: Distribution of Quality Scores in Red Wine Dataset

Observation: The majority of red wine samples are rated as quality 5 and 6, which together make up the bulk of the dataset. There is a sharp drop for higher quality scores (7 and 8), and very few samples with scores below 4.

Quality Distribution – White Wine

The same analysis was performed on the white wine dataset.

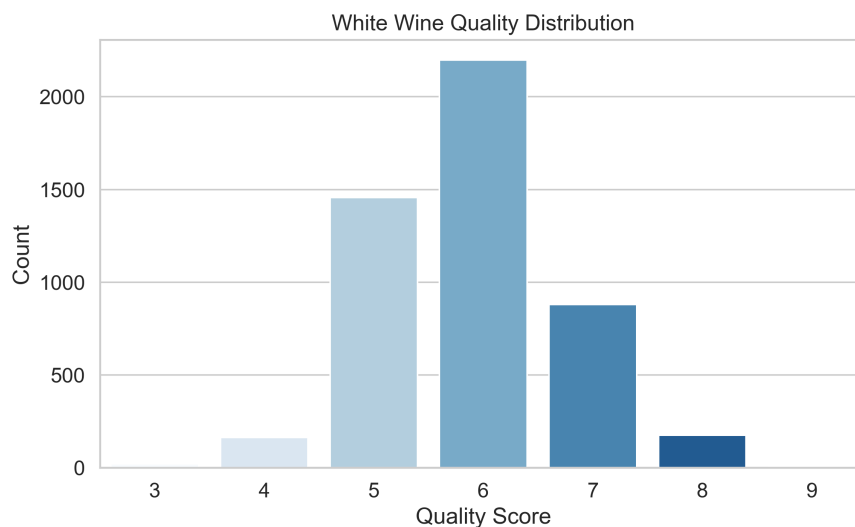


Figure 2: Distribution of Quality Scores in White Wine Dataset

Observation: Like red wines, most white wines fall into the 5–6 quality range. However, white wines show a broader spread — with more wines rated as 7 and 8 compared to red. There’s also a noticeable number of white wines rated as 4, which wasn’t as prominent in the red wine data.

Red vs. White Comparison

To better understand how red and white wines compare, both quality distributions were plotted together (see Figure 3). It is important to note that the white wine dataset contains over three times as many samples as the red wine dataset (4,898 vs. 1,599), which impacts the visual scale and interpretation.

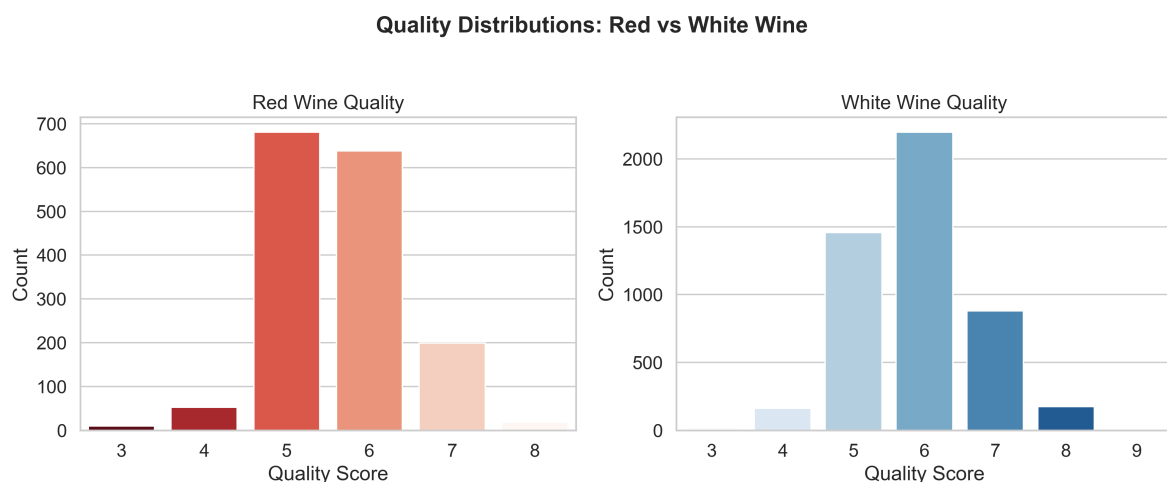


Figure 3: Side-by-Side Distribution of Red and White Wine Quality

Interpretation:

- **Red Wines:** Quality scores are tightly clustered around 5 and 6, with relatively few samples scoring outside that range. This suggests consistency, but also limited variation.
- **White Wines:** While also peaking at 5 and 6, the distribution is notably wider. A significant number of samples score 7, 8, or even 9 — which are absent from the red wine data. There are also more wines scoring 4.
- **Sample Size Consideration:** The broader spread in the white wine distribution is partly due to the **larger sample size**, which introduces more variability and makes higher and lower quality scores more visible.
- **Overall Implication:** While both types of wine tend to cluster around mid-level quality, white wine exhibits a **greater diversity** in perceived quality, which may point to differences in grape variety, fermentation control, or batch variation.

These early observations will guide further investigation into physicochemical factors that may explain why some wines — especially white — achieve higher sensory scores.

Task 1B: Alcohol Content and Wine Quality

Overview

Task 1B explores how the `alcohol` variable relates to perceived wine quality. A key idea in this analysis is that alcohol concentration — influenced by sugar and fermentation — can potentially explain differences in sensory quality ratings.

The alcohol values in both datasets were categorised into three levels: `low`, `mid`, and `high`, based on their mean and standard deviation:

- `low` $\hat{=}$ (mean - std)
- `mid` within 1 std of the mean
- `high` $\hat{=}$ (mean + std)

This discretisation results in a new variable, `alcohol_cat`, which is used to compare quality across different alcohol levels.

Alcohol vs Quality – Red Wine

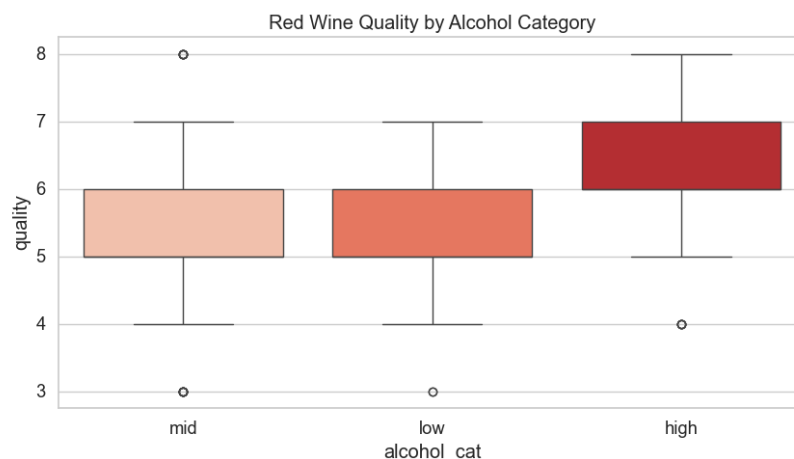


Figure 4: Red Wine Quality by Alcohol Category

Observation:

- Red wines with `high` alcohol content tend to have higher quality scores than those with `low` alcohol content.
- The distribution is clearly shifted upwards with increasing alcohol.
- Low alcohol wines rarely score above 6.

Alcohol vs Quality – White Wine

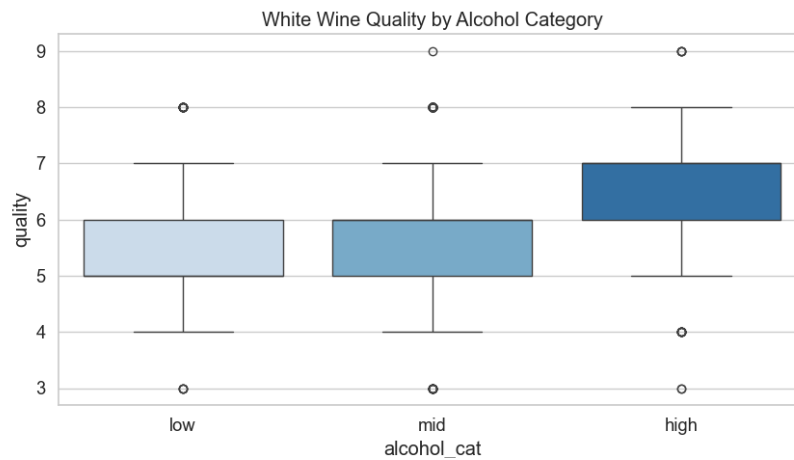


Figure 5: White Wine Quality by Alcohol Category

Observation:

- A similar trend is observed for white wine: higher alcohol wines tend to score better in quality.
- The difference between categories is subtler than in red wine but still visible.
- Low alcohol wines again perform worse overall.

Interpretation

- In both wine types, there is a clear positive relationship between alcohol content and quality rating.
- This does not imply causation — alcohol may be correlated with other contributing factors (e.g. fermentation duration, residual sugar, or acidity).
- However, `alcohol` appears to be a strong candidate for predictive modelling in future tasks.

This analysis reinforces the idea that alcohol content is a meaningful signal of perceived quality in wine production and should be further explored in conjunction with other chemical variables.